EAST WEST UNIVERSITY

Dhaka Drive Automobile Driving Training Institute

(A System Design and Analysis Project)

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1.**Introduction**

Purpose
Scope
Intended Audience and Reading Suggestions

2.**Usage scenario**

3.**System Features and Requirements**

Functional Requirements
Non-Functional Requirements

4.**Use case Diagram**

5.**Activity Diagram**

Login
Payment
Enrollment
Learning Material access

6.Class Diagram

7.**Data Flow Diagram**

Level 0 :
Level 1

8.**E-R Diagram**
9.**Sequence Diagram**

10.Project Snapshot
11.Limitations
12.Conclusion.

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for the **Dhaka Drive Automobile Driving Training Institute Management System**. This system will provide students with access to driving training schedules, course information, payments, and communication with instructors. It will also allow administrators to manage student records, course batches, payments, and exam results in a centralized, user-friendly platform.

1.2 Scope

The system is designed to serve as a **comprehensive digital solution** for managing all training-related activities of the driving institute. It will enable:

- Students to register, enroll in driving courses, view schedules, track progress, and pay fees.
- Instructors to manage training sessions, upload test results, and communicate with students.
- Administrators to oversee enrollment, vehicle allocation, exam scheduling, and certificate issuance.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- **Students & Trainees** – to understand how to use the portal for course enrollment, payments, and progress tracking.
- **Instructors** – to manage class schedules, training progress, and exam assessments.
- **Administrators** – to oversee student records, allocate vehicles, and maintain accounts.
- **Developers & Designers** – to implement the system as per requirements.
- **Testers** – to ensure all features function correctly.

2.Usage scenario

When a student enrolls in Dhaka Drive, the administrator creates a student profile with a unique ID. Students then activate their accounts using email/phone verification and set a password.

From their dashboard, students can:

- View available driving courses (car, motorcycle, professional license training).
- Book class schedules and track attendance.
- Pay fees online or check pending payments.
- Appear for theoretical and practical driving exams.
- Download certificates after course completion.

Instructors log in to:

- Manage their assigned classes.
- Record student attendance and progress.
- Upload test scores (theory & practical).

Administrators can:

- Manage courses, schedules, and vehicles.
- Assign instructors to students.
- Track payments, issue receipts, and approve certificates.

3. System Features and Requirements

3.1 Functional Requirements

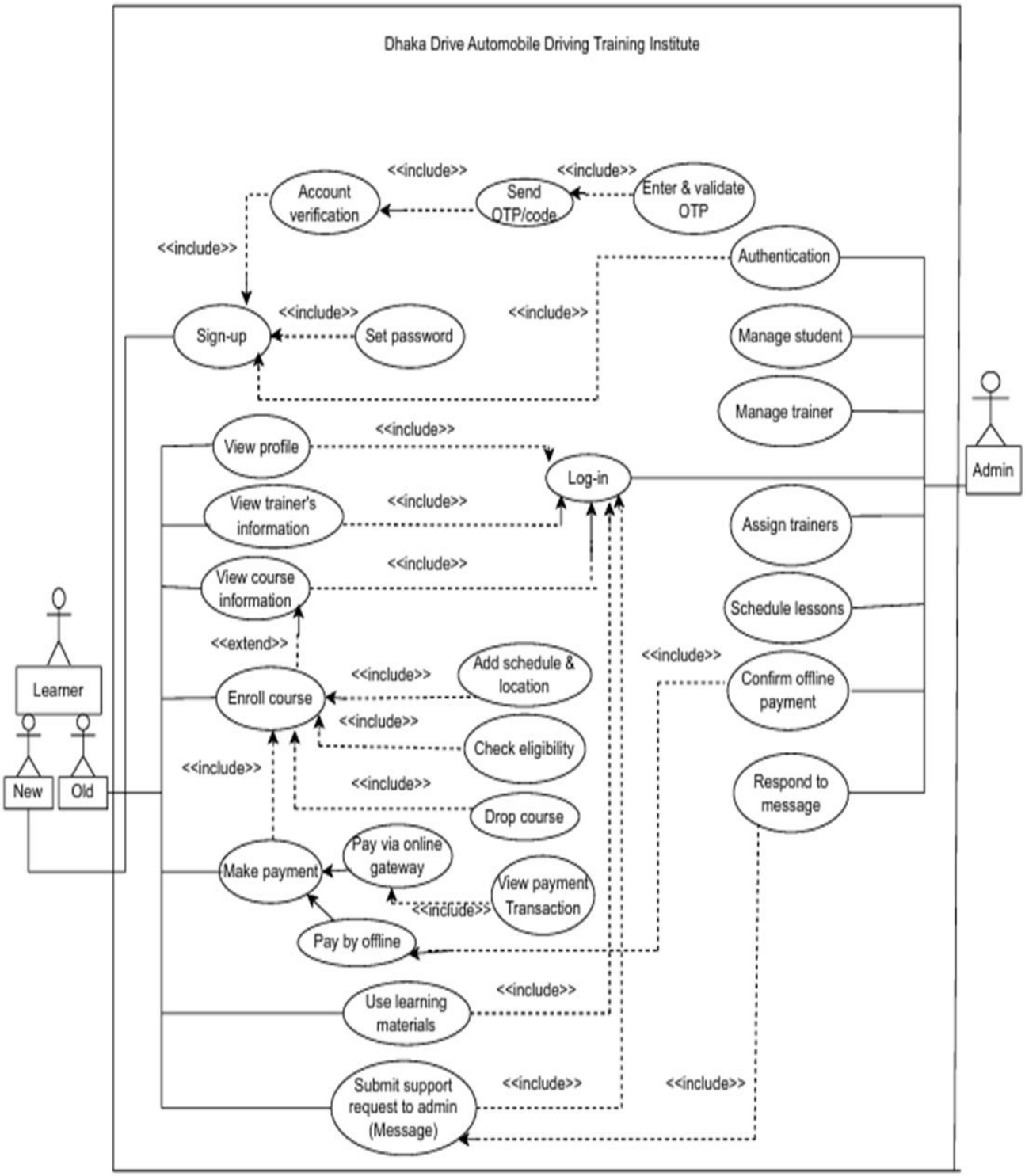
Stakeholder	Stakeholder Goal	Functional Requirement
Student	We want secure access to manage our training activities.	The system shall require students to activate their account and set a new password on first login.
	We want to ensure our personal data isn't altered without approval.	The system shall prevent students from changing their name, email, or phone number via the portal.
	We want to view all our course and personal information in one place.	The system shall allow students to view their profiles, including name, address, enrolled course, training status, and exam results.
	We want to stay informed about our training schedules.	The system shall allow students to view current courses, instructors, and training schedules.
	We want to monitor our training progress.	The system shall allow students to track progress in both theory and practical sessions.
	We want to manage our tuition and training fees.	The system shall allow students to view fee details, payment history, and outstanding balances.
	We want to book training sessions easily.	The system shall allow students to book or reschedule training sessions based on instructor availability.
	We want to contact instructors easily.	The system shall allow students to search and view instructor details (contact, schedule).
	We want to apply for	The system shall allow students

	license exams when eligible.	to register for driving exams after course completion.
	We want to download certificates.	The system shall allow students to download a completion certificate after successfully passing exams.
Instructor	We want to manage training batches.	The system shall allow instructors to schedule and update training sessions.
	We want to record student performance.	The system shall allow instructors to update student attendance, progress, and test results.
Admin	We want to onboard students securely.	The system shall allow admins to create accounts with unique IDs and credentials.
	We want to enforce policies and control access.	The system shall allow admins to block, suspend, or dismiss student accounts.
	We want to maintain accurate student and vehicle records.	The system shall allow admins to update student and vehicle information when needed.
System Admin	We want to ensure reliable system performance.	The system shall allow administrators to monitor and maintain system functionality.

3.2 Non-Functional Requirements

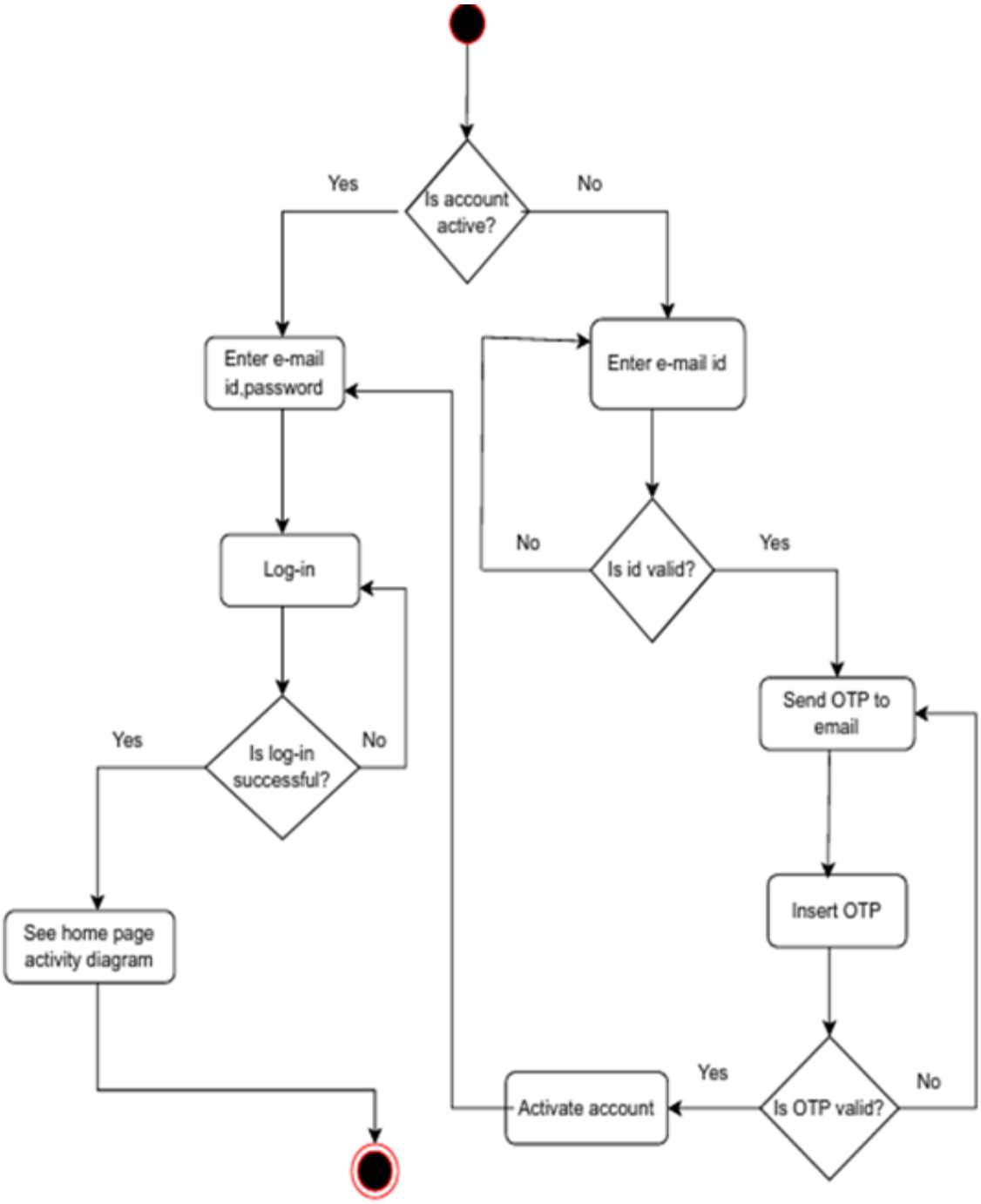
Stakeholder	Stakeholder Goal	Non-Functional Requirement
Student	We want to access the system from multiple devices.	The system shall support simultaneous logins from PC and mobile.
	We want quick and smooth booking during peak times.	The system shall ensure fast and responsive booking functions.
	We want an easy-to-use interface.	The system shall offer a user-friendly and intuitive interface.
	We want our feedback to remain anonymous.	The system shall guarantee anonymity in the feedback system.
Admin	We want to protect student and training data.	The system shall ensure secure storage and handling of student information.
System Admin	We want high availability during peak periods.	The system shall maintain 99.9% uptime during enrollment and exam registration periods.
	We want scalability to support many users.	The system shall be scalable to handle simultaneous user access.
	We want strong data security.	The system shall use encryption and access controls to protect sensitive data.

4. Use case Diagram

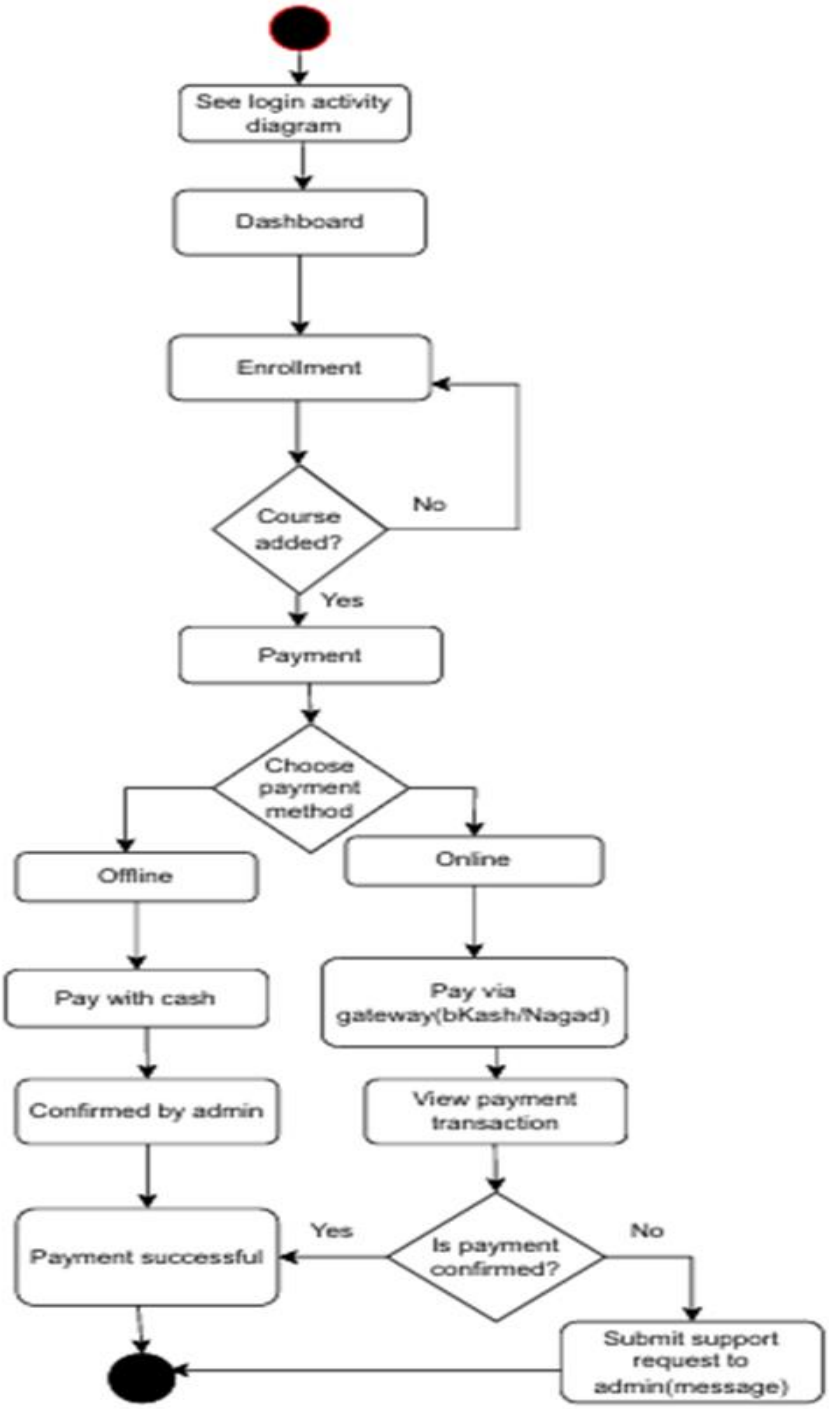


5. Activity Diagram

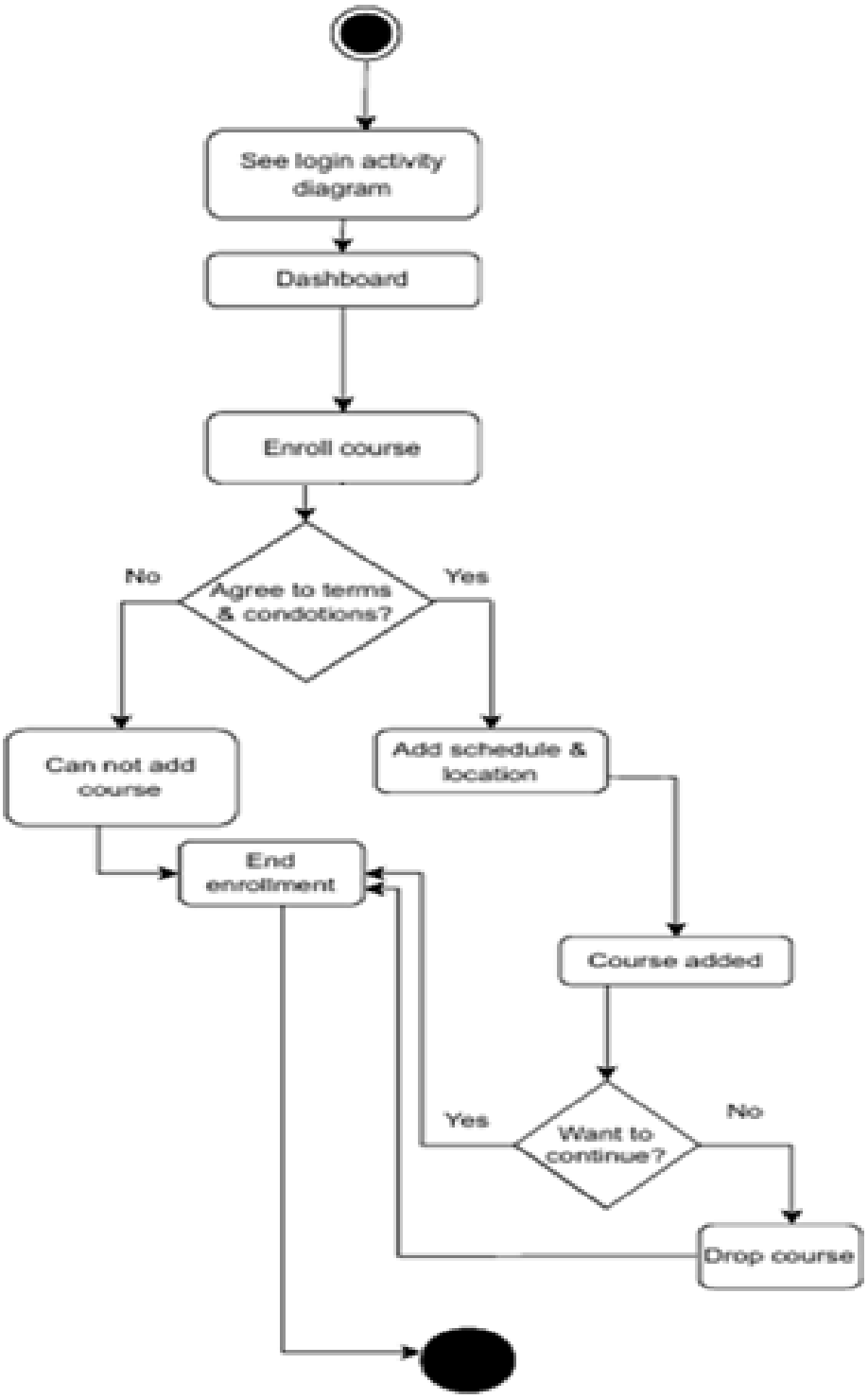
Log-in:



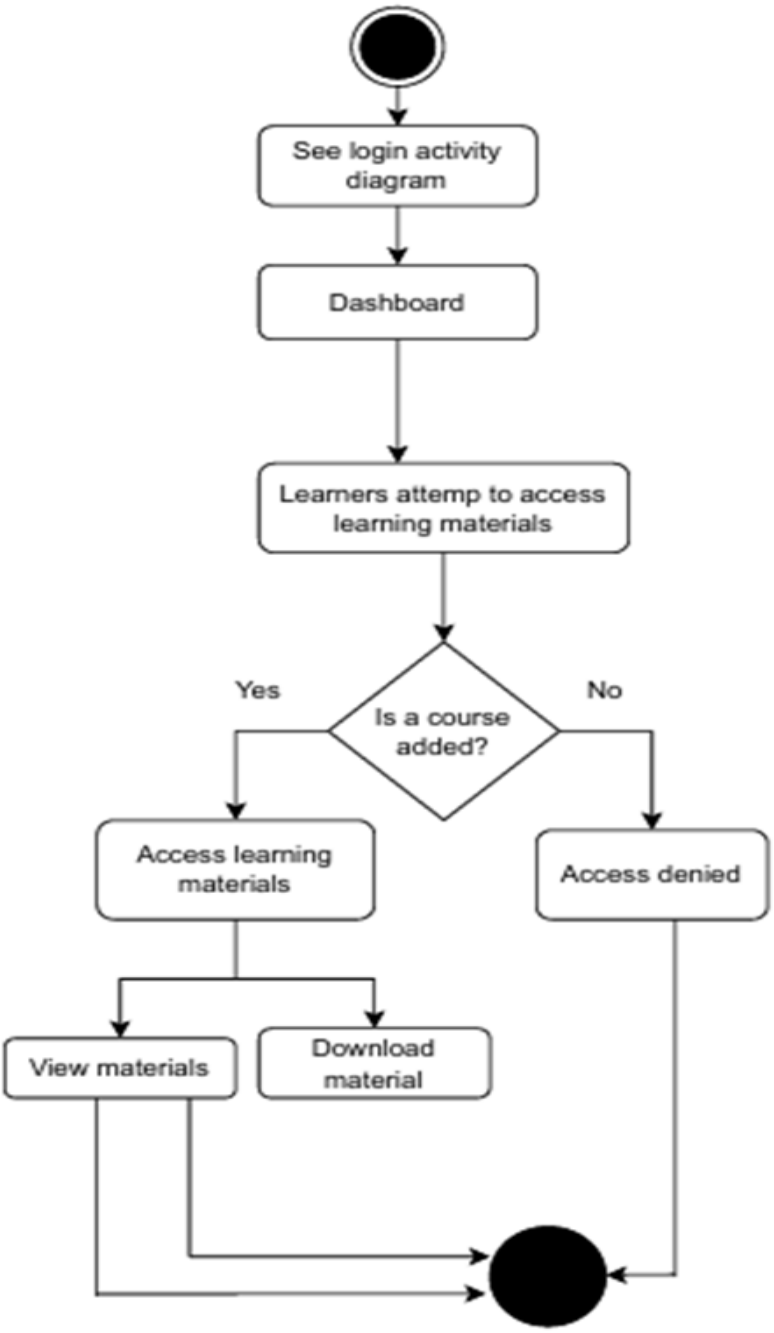
Payment:


See login activity diagram
Dashboard
Enrollment
Course added?
No
Yes
Payment
Choose payment method
Offline
Online
Pay with cash
Pay via gateway(bKash/Nagad)
Confirmed by admin
View payment transaction
Payment successful
Yes
Is payment confirmed?
No
Submit support request to admin(message)

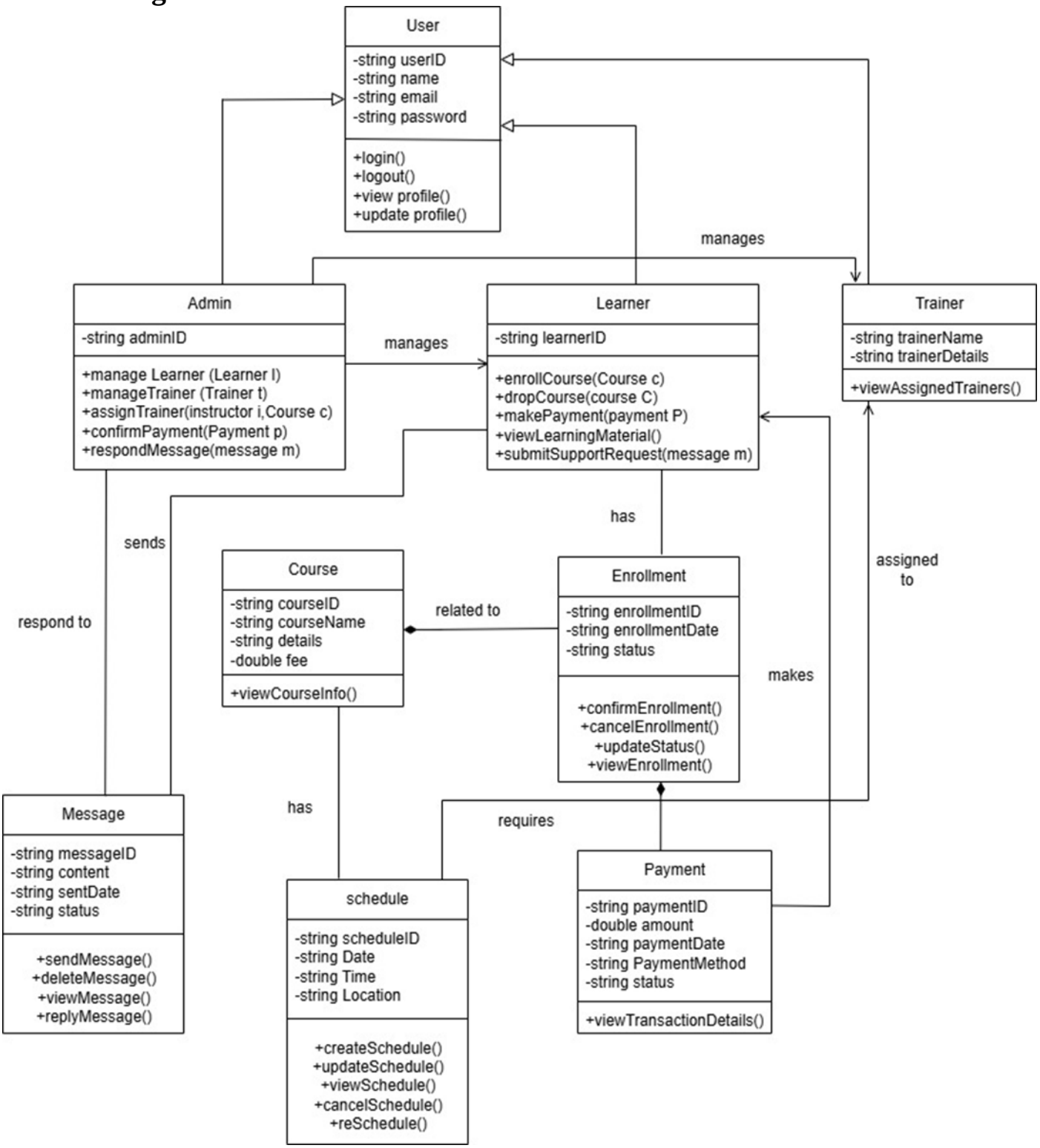
Enrollment:


See login activity diagram
Dashboard
Enroll course
Agree to terms & condotions?
No
Yes
Can not add course
Add schedule & location
End enrollment
Course added
Want to continue?
Yes
No
Drop course

Learning Material Access:

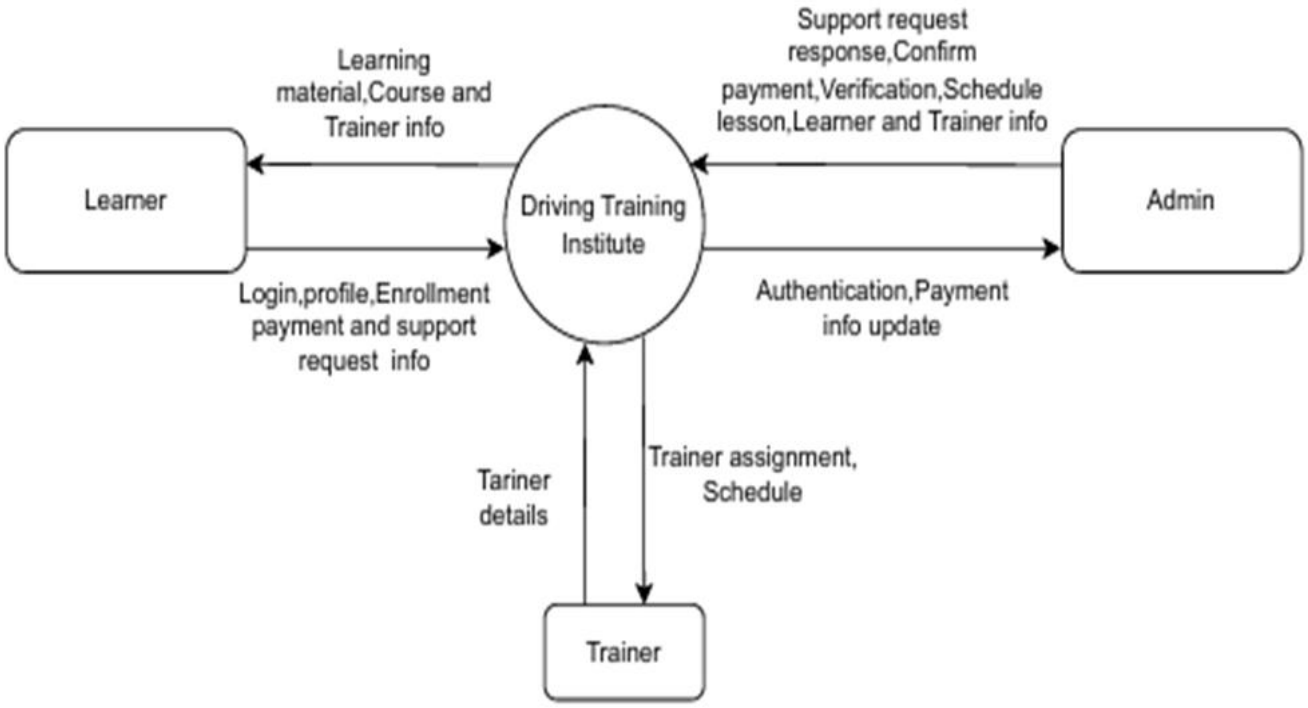


6. Class Diagram:

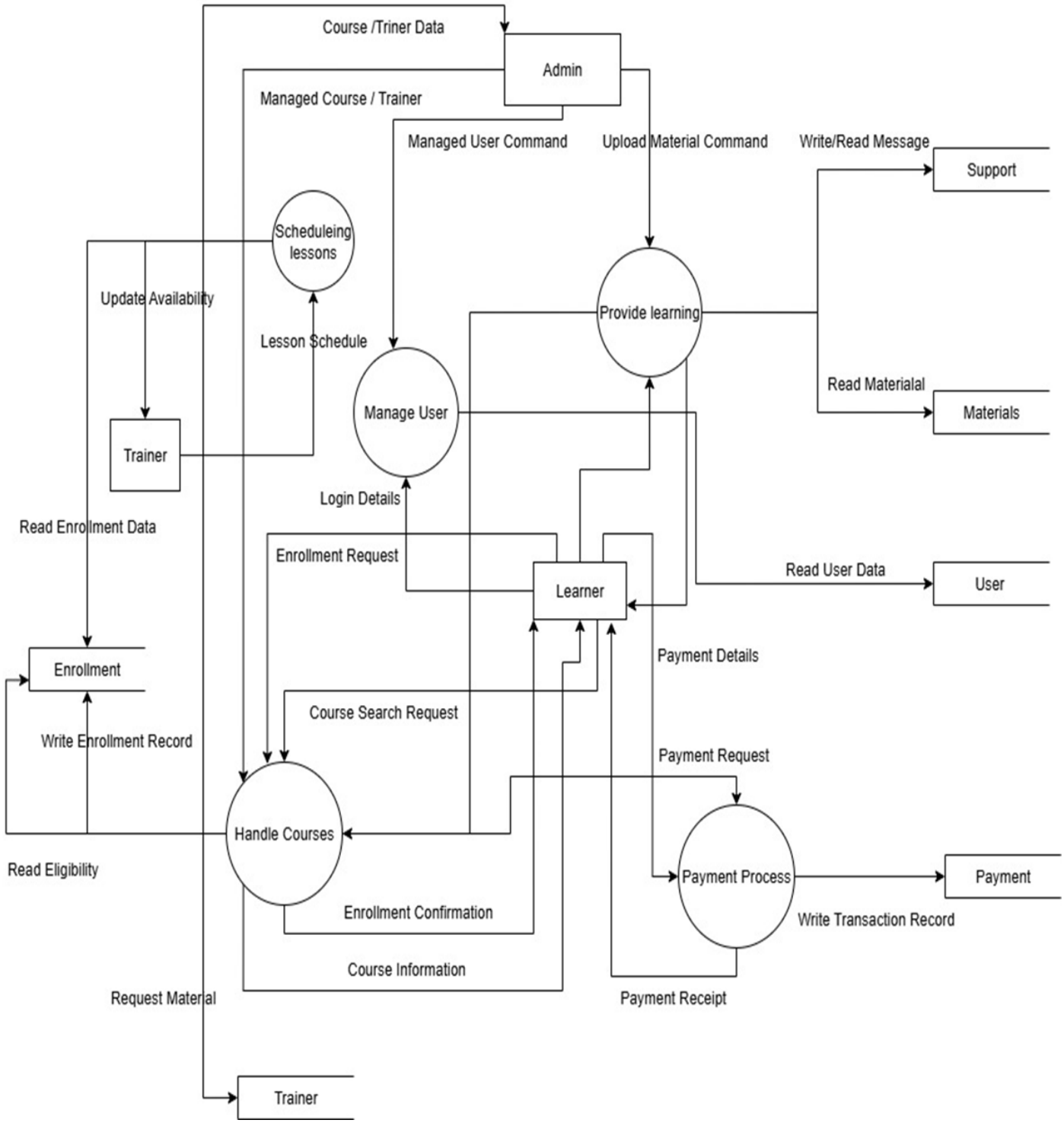


7. Data Flow Diagram:

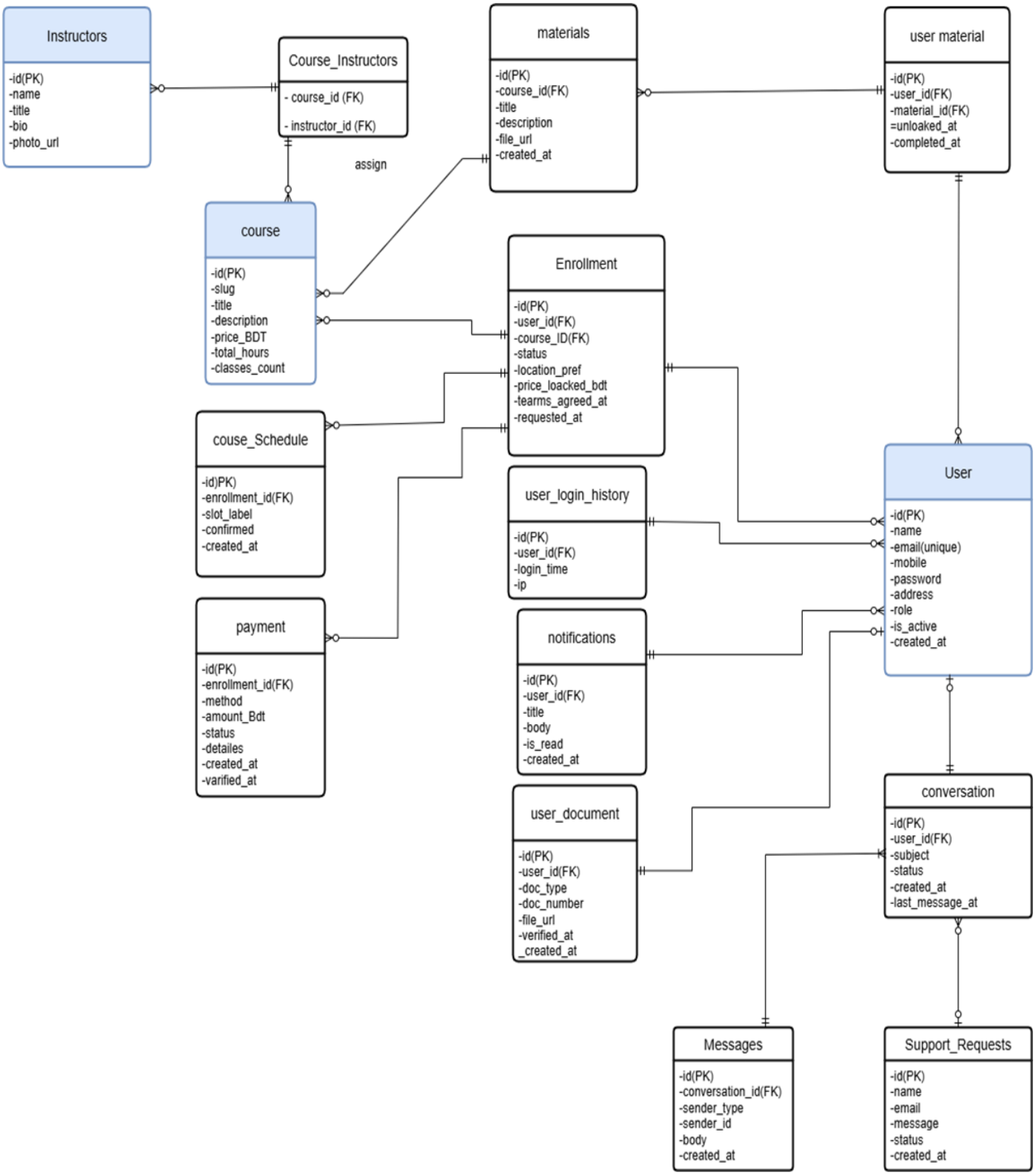
Level 0:



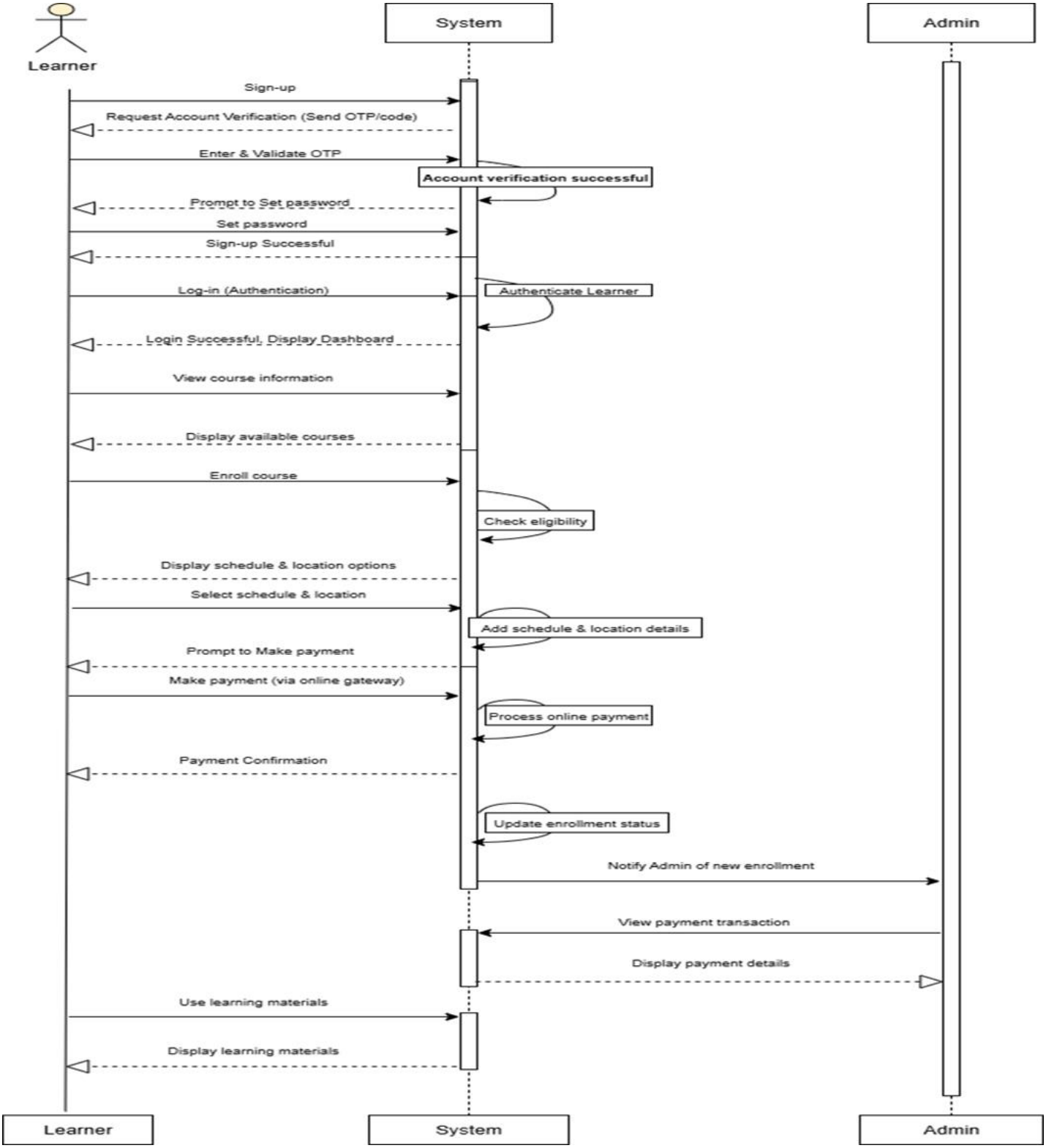
Level 1:


Course /Triner Data
Admin
Managed Course / Trainer
Managed User Command
Upload Material Command
Write/Read Message
Support
Scheduleing lessons
Update Availability
Lesson Schedule
Provide learning
Read Materialal
Materials
Manage User
Trainer
Login Details
Read Enrollment Data
Enrollment Request
Learner
Read User Data
User
Enrollment
Payment Details
Course Search Request
Write Enrollment Record
Payment Request
Handle Courses
Read Eligibility
Payment Process
Payment
Enrollment Confirmation
Write Transaction Record
Course Information
Request Material
Payment Receipt
Trainer

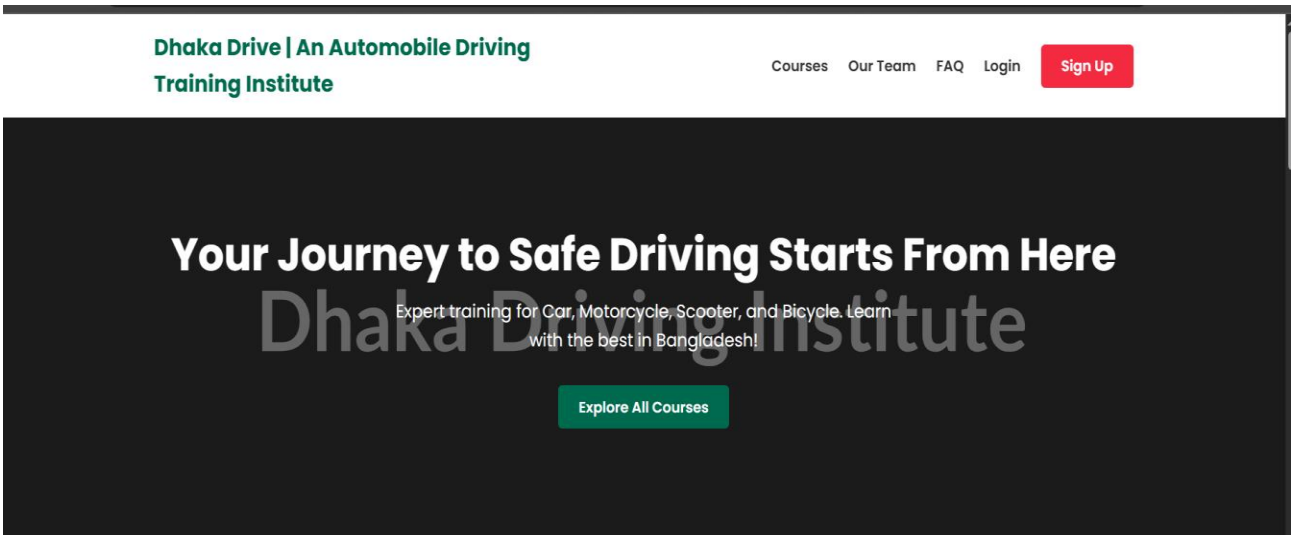
8. E-R Diagram:

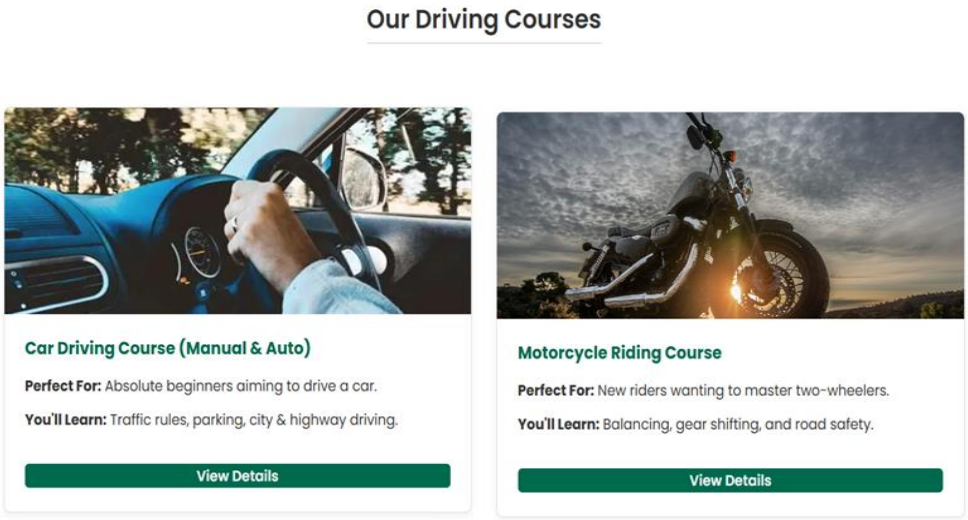

Instructors
-id(PK)
-name
-title
-bio
-photo_url
Course_Instructors
- course_id (FK)
- instructor_id (FK)
assign
materials
-id(PK)
-course_id(FK)
-title
-description
-file_url
-created_at
user material
-id(PK)
-user_id(FK)
-material_id(FK)
=unloaked_at
-completed_at
course
-id(PK)
-slug
-title
-description
-price_BDT
-total_hours
-classes_count
Enrollment
-id(PK)
-user_id(FK)
-course_ID(FK)
-status
-location_pref
-price_loacked_bdt
-tearms_agreed_at
-requested_at
couse_Schedule
-id)PK)
-enrollment_id(FK)
-slot_label
-confirmed
-created_at
user_login_history
-id(PK)
-user_id(FK)
-login_time
-ip
User
-id(PK)
-name
-email(unique)
-mobile
-password
-address
-role
-is_active
-created_at
payment
-id(PK)
-enrollment_id(FK)
-method
-amount_Bdt
-status
-detailes
-created_at
-varified_at
notifications
-id(PK)
-user_id(FK)
-title
-body
-is_read
-created_at
user_document
-id(PK)
-user_id(FK)
-doc_type
-doc_number
-file_url
-verified_at
_created_at
conversation
-id(PK)
-user_id(FK)
-subject
-status
-created_at
-last_message_at
Messages
-id(PK)
-conversation_id(FK)
-sender_type
-sender_id
-body
-created_at
Support_Requests
-id(PK)
-name
-email
-message
-status
-created_at

9. Sequence Diagram:



10. Project snapshot:





What Our Students Say

"Dhaka Drive made learning so easy! My instructor was incredibly patient. I passed my test on the first try. Highly recommended!"

- Sadia Islam, Car Driving Student

"I was always scared of riding a motorcycle in Dhaka traffic. The instructors here gave me the confidence and skills I needed. Thank you!"

- Rifat Ahmed, Motorcycle Student

"The license assistance service is a lifesaver. They guided me through every step of the BRTA paperwork, which saved me a lot of hassle."

- Nabil Haque, Car Driving Student

Get Started Today!

Call us or visit our office to book your spot. Your driving adventure awaits!

Phone: +880 1712-345678
Email: info@dhakadrive.com.bd
Location: House 123, Road 4, Dhanmondi, Dhaka-1205, Bangladesh

Call Us Now

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Motorcycle Riding

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Master the art of two-wheeling with our comprehensive motorcycle course. Ideal for new riders, we focus on building your confidence, control, and road safety awareness to handle Dhaka's dynamic traffic.

☐ I have read and agree to the Terms and Conditions.

Request to Enroll

11. Limitations:

1. **Limited User Base:** The system is designed primarily for students and staff of Dhaka Drive, which may limit its scalability for other driving institutes without customization.
2. **Internet Dependency:** Some features, such as online registration or progress tracking, require a stable internet connection, which may not always be available to all users.
3. **Hardware Constraints:** The software may face performance issues on low-end devices or older computers used by staff or students.
4. **Manual Data Entry Errors:** Although the system streamlines operations, errors in manual data entry (like attendance or test results) may still occur.
5. **Limited Real-Time Feedback:** The system may not fully provide real-time driving performance analysis for learners; practical evaluations are still dependent on instructors.

12. Conclusion:

The **Dhaka Drive Automobile Driving Training Institute** project successfully automates key operations of a driving school, including student registration, course management, scheduling, and performance tracking. It provides a user-friendly interface for both students and staff, reducing paperwork, minimizing manual errors, and improving overall efficiency. Despite limitations like dependency on internet connectivity and hardware constraints, the system serves as an effective solution for managing driving training operations digitally. With future enhancements, such as real-time driving performance feedback and mobile app integration, the system can become even more comprehensive and accessible.